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SCSR1013 DIGITAL LOGIC

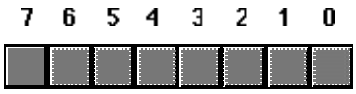
MODULE 2b: DATA ORGANIZATION (CODES)

2023/2024-1

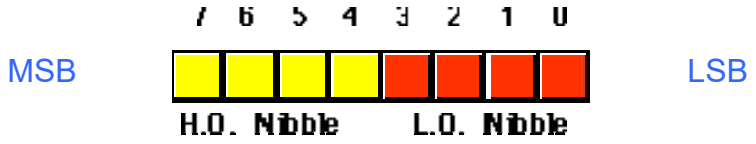
FACULTY OF COMPUTING

Data Organization

- A value may take an arbitrary number of bits.
- Common collections are single bits
  - smallest "unit" of data on a binary computer is a single bit
  - groups of four bits called nibbles
  - groups of eight bits called bytes
  - groups of 16 bits called words
- The bits in a byte are normally numbered from zero to seven.



- Bit 0 is the low order bit (rightmost) or least significant bit (LSB) bit 7 is the high order bit (leftmost) or most significant bit (MSB) of the byte.
  Note **1 byte** also contains exactly **2 nibbles**:



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Nibbles

- 4 bits



- **Major uses**:
  - BCD (Binary Coded Decimal)
  - Hexadecimal numbers

Example: 0111, 1011 and 1111.

7_{16}, B_{16}, F_{16}

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Bytes



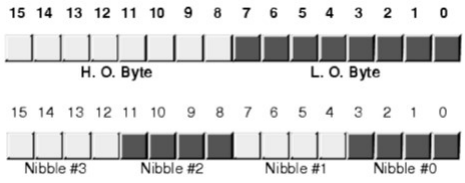
- 8 bits
- Total values: $2^8 = 256$

- **Major uses**:
  - Numeric values (0 … 2^8-1 = 0 … 255)
  - Signed numbers: (-128 to +127)

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Word

- 16 bits = 2 bytes
- Bit 0 to 15
- **Total values**:
  - $2^{16} = 65{,}536$



- **Major uses of word**:
  - signed integer (-32,768 … +32,767)
  - unsigned integer (0 … 2^{16}-1) = 0 … 65,535)
    UNICODE characters

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What are codes?

- Code is a representation of information generated by following a certain rules.
- In general, we need code because:
  - Code is unique
  - Codes are easy to process
  - Code is easy to represent
  - Codes enable communication in place where ordinary spoken or written language is difficult or impossible, eg Morse Code
- Due to this, code can simplify the process (such as manipulation and arithmetic operations) of the information in the digital system.

We will learn:

i. BCD codes
ii. Gray Codes
iii. ASCII codes
iv. Parity codes/bit

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Binary Coded Decimal (BCD)

- BCD is a way to express each of the decimal digits with a binary code.
- There are only 10 code groups in the BCD system, one for every digit (0000 – 1001)

Decimal	BCD	Decimal	BCD
0	0000	5	0101
1	0001	6	0110
2	0010	7	0111
3	0011	8	1000
4	0100	9	1001

Invalid codes are 1010, 1011, 1100, 1101, 1110, 1111

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Example 1: Convert 3245 to BCD

3 2 4 5

3245 = 0011 0010 0100 0101

Example 2: Convert 7848 to BCD

7 8 4 8

7848 = 0111 1000 0100 1000

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Gray codes

Gray Codes

Electromechanical switches



Basics Toggle Limit

- Designed to prevent **false output** from electromechanical switches.
- Are widely used to facilitate **error correction** in digital communications such as digital terrestrial television and some cable TV systems.
- In modern digital communications, Gray codes play an important role in error correction.
- It is arranged so that every transition from one value to the next value involves only one bit change.
- Sometimes referred to as reflected binary, l because the first eight values compare with those of the last 8 values, but in reverse order.

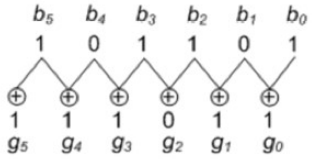
reflected binary

compare

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Conversion of Gray Code ←→ Binary

To convert **binary → Gray code**



where b_{i+1}, b_i, b_{i-1}, …, b_0 is the binary number while g_{i+1}, g_i, g_{i-1}, …, g_0 gray code.

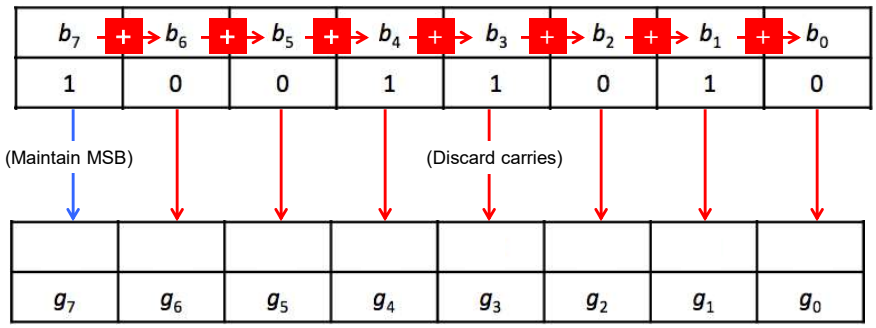
To convert **Gray code → binary:**

- $b_i = g_i$ if no. of 1's preceding g_i is even
- $b_i = \overline{g_i}$ if no. of 1's preceding g_i is odd

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Example:

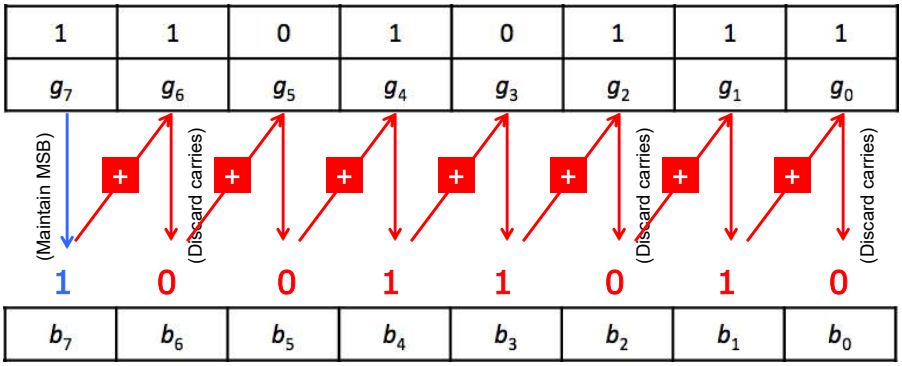
Convert 100110 to its equivalent gray code value



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Example:

Convert the Gray code 11010111 to binary.



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Extra

Exercise 2b.1:

Convert 11010100_2 to its equivalent gray code value.

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Parity Code

- Parity bit used for bit error detection
  - Even parity – total number of 1s even
  - Odd parity – total number of 1s odd
- Parity bit is append to the code at the leftmost position (MSB) .

Parity bit

0

0

A parity bit is a bit that is added to ensure that the number of bits with value of 1's in a given set of bits is always even or odd. Parity bits are used as the simplest error detecting code.

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Extra

Examples:

1 1 0 1 0 0 1 1 1

Even Parity bit

0 1 0 1 0 0 1 1 1

Odd Parity bit

Number 1s	Even Parity	Odd Parity
Even	0	1
Odd	1	0

(Remember these basic rule)

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7 bits of data (number of 1s)	8 bits including parity	
	even	odd
0000000 (0)	00000000	10000000
1010001 (3)	11010001	01010001
1101001 (4)	01101001	11101001
1111111 (7)	11111111	01111111

Parity bit

Resource: http://en.wikipedia.org/wiki/Parity_bit

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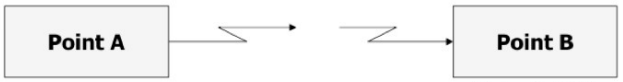
- Example: Calculate the parity bit for the codes below.

Code	Number of 1s	Even/Odd	Even Parity	Odd Parity
110010	3	Odd	**1** 110010	**0** 110010
101110	4	Even	**0** 101110	**1** 101110
101000	2	Even	**0** 101000	**1** 101000
110111	5	Odd	**1** 110111	**0** 110111
111111	6	Even	0 111111	1 111111
100000	1	Odd	1 100000	0 100000

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Error Detection by Parity Checking

- Assume that data = 0101
- It uses even parity.
- Therefore the appended parity bit is 0.
- The data with parity bit: 0 0101
- The data is transmitted.
- The data is received as 00001 → odd no. of 1, not even!!



Data: 0 0101
(even parity)

Data: 0 0001
(even parity)
but does not conform
to even parity

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Module 2

American Standard Code for Information Interchange (ASCII)

- It has 128 characters and symbols represented in 7-bit binary code
- Example :
- A = 1000001_2
- a = 1100001_2
- A parity bit is added so that the total number of bits is 8 → a byte.

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Decimal	Hex	ASCII	Decimal	Hex	ASCII	Decimal	Hex	ASCII	Decimal	Hex	ASCII
0	00	NUL	32	20	(blank)	64	40	@	96	60	`
1	01	SOH	33	21	!	65	41	A	97	61	a
2	02	STX	34	22	"	66	42	B	98	62	b
3	03	ETX	35	23	#	67	43	C	99	63	c
4	04	EOT	36	24	$	68	44	D	100	64	d
5	05	ENQ	37	25	%	69	45	E	101	65	e
6	06	ACK	38	26	&	70	46	F	102	66	f
7	07	BEL	39	27	'	71	47	G	103	67	g
8	08	BS	40	28	(	72	48	H	104	68	h
9	09	HT	41	29	)	73	49	I	105	69	i
10	0A	LF	42	2A	*	74	4A	J	106	6A	j
11	0B	VT	43	2B	+	75	4B	K	107	6B	k
12	0C	FF	44	2C	,	76	4C	L	108	6C	l
13	0D	CR	45	2D	-	77	4D	M	109	6D	m
14	0E	SO	46	2E	.	78	4E	N	110	6E	n
15	0F	SI	47	2F	/	79	4F	O	111	6F	o
16	10	DLE	48	30	0	80	50	P	112	70	p
17	11	DC1	49	31	1	81	51	Q	113	71	q
18	12	DC2	50	32	2	82	52	R	114	72	r
19	13	DC3	51	33	3	83	53	S	115	73	s
20	14	DC4	52	34	4	84	54	T	116	74	t
21	15	NAK	53	35	5	85	55	U	117	75	u
22	16	SYN	54	36	6	86	56	V	118	76	v
23	17	ETB	55	37	7	87	57	W	119	77	w
24	18	CAN	56	38	8	88	58	X	120	78	x
25	19	EM	57	39	9	89	59	Y	121	79	y
26	1A	SUB	58	3A	:	90	5A	Z	122	7A	z
27	1B	ESC	59	3B	;	91	5B	[	123	7B	{
28	1C	FS	60	3C	<	92	5C	\	124	7C	|
29	1D	GS	61	3D	=	93	5D	]	125	7D	}
30	1E	RS	62	3E	>	94	5E	^	126	7E	~
31	1F	US	63	3F	?	95	5F	_	127	7F	(delete)

Resource: http://www.s4a.us/support/cimco_faq.htm

Extra

7-bits binary$_2$ ASCII code

Examples:

$B_7b_6b_5$ $b_4b_3b_2b_1$

110 1101

is represent as '**m**'

b_4 b_3 b_2 b_1	Row ↓ / Column → 0	1	2	3	4	5	6	7
	b_7 b_6 b_5 → 0 0 0	0 0 1	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
0 0 0 0	0	NUL	DLE	SP	0	@	P	`	p
0 0 0 1	1	SOH	DC1	!	1	A	Q	a	q
0 0 1 0	2	STX	DC2	"	2	B	R	b	r
0 0 1 1	3	ETX	DC3	#	3	C	S	c	s
0 1 0 0	4	EOT	DC4	$	4	D	T	d	t
0 1 0 1	5	ENQ	NAK	%	5	E	U	e	u
0 1 1 0	6	ACK	SYN	&	6	F	V	f	v
0 1 1 1	7	BEL	ETB	'	7	G	W	g	w
1 0 0 0	8	BS	CAN	(	8	H	X	h	x
1 0 0 1	9	HT	EM	)	9	I	Y	i	y
1 0 1 0	10	LF	SUB	*	:	J	Z	j	z
1 0 1 1	11	VT	ESC	+	;	K	[	k	{
1 1 0 0	12	FF	FC	,	<	L	\	l	|
1 1 0 1	13	CR	GS	-	=	M	]	m	}
1 1 1 0	14	SO	RS	.	>	N	^	n	~
1 1 1 1	15	SI	US	/	?	O	_	o	DEL

Resource: http://en.wikipedia.org/wiki/File:ASCII_Code_Chart-Quick_ref_card.png

Exercise 2b.2:

Convert the string SCR1013 to its ASCII hexadecimal value.

SCR1013 = 53 43 52 31 30 31 33

By using even parity coding, calculate the parity bit and insert this bit at the MSB position. Recalculate the ASCII value in its hexadecimal representation.

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Extra

Exercise 2b.3:

Given a string UTM1435h.

a) Convert the string to its ASCII hexadecimal value.

b) Calculate the odd parity bit and insert as MSB.

a) Recalculate the ASCII value in hexadecimal.

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Number 1s	Even Parity	Odd Parity
Even	0	1
Odd	1	0

Extra

Character (ASCII)	ASCII (Hex)	Binary	Odd parity bit + Binary	New ASCII (Hex)
U	55	101 0101	1101 0101	D5
T	54	101 0100	0101 0100	54
M	4D	100 1101	1100 1101	CD
1	31	011 0001	0011 0001	31
4	34	011 0100	0011 0100	34
3	33	011 0011	1011 0011	B3
5	35	011 0101	1011 0101	B5
h	68	110 1000	0110 1000	68

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Decimal	Hex	ASCII	Decimal	Hex	ASCII	Decimal	Hex	ASCII	Decimal	Hex	ASCII
0	00	NUL	32	20	(blank)	64	40	@	96	60	`
1	01	SOH	33	21	!	65	41	A	97	61	a
2	02	STX	34	22	"	66	42	B	98	62	b
3	03	ETX	35	23	#	67	43	C	99	63	c
4	04	EOT	36	24	$	68	44	D	100	64	d
5	05	ENQ	37	25	%	69	45	E	101	65	e
6	06	ACK	38	26	&	70	46	F	102	66	f
7	07	BEL	39	27	'	71	47	G	103	67	g
8	08	BS	40	28	(	72	48	H	104	68	h
9	09	HT	41	29	)	73	49	I	105	69	i
10	0A	LF	42	2A	*	74	4A	J	106	6A	j
11	0B	VT	43	2B	+	75	4B	K	107	6B	k
12	0C	FF	44	2C	,	76	4C	L	108	6C	l
13	0D	CR	45	2D	-	77	4D	M	109	6D	m
14	0E	SO	46	2E	.	78	4E	N	110	6E	n
15	0F	SI	47	2F	/	79	4F	O	111	6F	o
16	10	DLE	48	30	0	80	50	P	112	70	p
17	11	DC1	49	31	1	81	51	Q	113	71	q
18	12	DC2	50	32	2	82	52	R	114	72	r
19	13	DC3	51	33	3	83	53	S	115	73	s
20	14	DC4	52	34	4	84	54	T	116	74	t
21	15	NAK	53	35	5	85	55	U	117	75	u
22	16	SYN	54	36	6	86	56	V	118	76	v
23	17	ETB	55	37	7	87	57	W	119	77	w
24	18	CAN	56	38	8	88	58	X	120	78	x
25	19	EM	57	39	9	89	59	Y	121	79	y
26	1A	SUB	58	3A	:	90	5A	Z	122	7A	z
27	1B	ESC	59	3B	;	91	5B	[	123	7B	{
28	1C	FS	60	3C	<	92	5C	\	124	7C	|
29	1D	GS	61	3D	=	93	5D	]	125	7D	}
30	1E	RS	62	3E	>	94	5E	^	126	7E	~
31	1F	US	63	3F	?	95	5F	_	127	7F	(delete)

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